Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.24b)

Metric

$$ds^2 = -a^2 \mathrm{d}t \otimes \mathrm{d}t + a^2 \mathrm{d}\chi \otimes \mathrm{d}\chi + a^2 \sin(\chi)^2 \, \mathrm{d}\theta \otimes \mathrm{d}\theta + a^2 \sin(\chi)^2 \sin(\theta)^2 \, \mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (t, \chi, \theta, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2}\sqrt{2}a\right) dt + \left(\frac{1}{2}\sqrt{2}a\right) d\chi$$

$$n = \left(\frac{1}{2}\sqrt{2}a\right) dt + \left(-\frac{1}{2}\sqrt{2}a\right) d\chi$$

$$m = \left(\frac{1}{2}\sqrt{2}a\sin(\chi)\right) d\theta + \left(\frac{1}{2}i\sqrt{2}a\sin(\chi)\sin(\theta)\right) d\phi$$

$$\overline{m} = \left(\frac{1}{2}\sqrt{2}a\sin(\chi)\right) d\theta + \left(-\frac{1}{2}i\sqrt{2}a\sin(\chi)\sin(\theta)\right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\,a}\right) dt + \left(\frac{\sqrt{2}}{2\,a}\right) d\chi$$

$$n = \left(-\frac{\sqrt{2}}{2\,a}\right)\,dt + \left(-\frac{\sqrt{2}}{2\,a}\right)\,d\chi$$

$$m = \left(\frac{\sqrt{2}}{2\,a\sin\left(\chi\right)}\right)\,d\theta + \left(\frac{i\,\sqrt{2}}{2\,a\sin\left(\chi\right)\sin\left(\theta\right)}\right)\,d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2\,a\sin\left(\chi\right)}\right)\,d\theta + \left(-\frac{i\,\sqrt{2}}{2\,a\sin\left(\chi\right)\sin\left(\theta\right)}\right)\,d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\,a}\,\partial_t + \frac{\sqrt{2}}{2\,a}\,\partial_\chi$$

$$\Delta = -\frac{\sqrt{2}}{2\,a}\,\partial_t - \frac{\sqrt{2}}{2\,a}\,\partial_\chi$$

$$\delta = \frac{\sqrt{2}}{2\,a\sin\left(\chi\right)}\,\partial_\theta + \frac{i\,\sqrt{2}}{2\,a\sin\left(\chi\right)\sin\left(\theta\right)}\,\partial_\phi$$

$$\overline{\delta} = \frac{\sqrt{2}}{2\,a\sin\left(\chi\right)}\,\partial_\theta - \frac{i\,\sqrt{2}}{2\,a\sin\left(\chi\right)\sin\left(\theta\right)}\,\partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}\cos\left(\chi\right)}{2\,a\sin\left(\chi\right)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\cos(\chi)}{2\,a\sin(\chi)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cos(\theta)}{4\,a\sin(\chi)\sin(\theta)}$$

$$\beta = \frac{\sqrt{2}\cos(\theta)}{4\,a\sin(\chi)\sin(\theta)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\Phi_{00} = \frac{1}{2\,a^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{4\,a^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{1}{2\,a^2}$$

$$\Lambda = \frac{1}{4\,a^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$\Psi_4 = 0$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"